

AI-powered bioinformatic analysis using R, #7

Data Visualization III — Overview of Omics Data & Visualization

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Section 1

Introduction

The Omics Revolution

Biology has entered the **big data era**. Modern high-throughput technologies generate vast amounts of data across multiple molecular layers:

- **Genomics** — DNA, the blueprint of life
- **Transcriptomics** — RNA, what genes are expressed
- **Proteomics** — Proteins, the molecular machines
- **Metabolomics** — Small molecules, the metabolic output
- **Epigenomics** — DNA modifications and chromatin structure
- **Microbiomics** — Communities of microorganisms

Common goal: turn raw data into biological insights through analysis and visualization.

Today's Goal

This talk provides an **overview** of:

- ① Major bioinformatics data types and their characteristics
- ② Common visualization tasks for each data type
- ③ Key R packages and tools used
- ④ The typical workflow: data → analysis → visualization → interpretation

In future courses, we will dive into each data type with **real-world datasets**, covering complete workflows from data analysis to visualization and results interpretation.

Section 2

Genomics Data

What is Genomics?

Genomics studies the complete set of DNA in an organism — the genome.

Typical data:

- **Whole Genome Sequencing (WGS)**: entire genome at base-pair resolution
- **Whole Exome Sequencing (WES)**: protein-coding regions only
- **Targeted sequencing**: specific genes or regions
- **GWAS**: genome-wide association studies linking variants to traits

Genomics: Data & Visualization

Data Type	Visualization Task	R Tools
Somatic mutations	Mutation landscape / oncoplot	maftools
Somatic mutations	Mutation signature spectrum	maftools, MutationalPatterns
Copy number variations	CNV heatmap, genome-wide profile	ComplexHeatmap, ggplot2
SNP / GWAS	Manhattan plot	qqman
SNP / GWAS	QQ plot (p-value distribution)	qqman
Structural variants	Circos plot, genome rearrangement	circlize, karyoploteR
Prokaryotic genomes	Gene structure / operon map	gggenes
Genome alignment	Syntenic dot plot	ggplot2, gggenomes

Genomics: Example Figures

- **Oncoplot:** rows = genes, columns = patients, colors = mutation types. Shows co-mutation patterns and mutation burden across a cancer cohort.
- **Manhattan plot:** x-axis = genomic position (by chromosome), y-axis = $-\log_{10}(p)$. Peaks above the significance threshold indicate trait-associated loci.
- **Circos plot:** circular genome view integrating multiple tracks (genes, CNV, SNPs, repeats) in one figure.

Section 3

Transcriptomics Data

What is Transcriptomics?

Transcriptomics measures RNA transcripts — revealing which genes are active under specific conditions.

Typical data:

- **Bulk RNA-seq**: average expression across millions of cells
- **Single-cell RNA-seq (scRNA-seq)**: expression in individual cells
- **Microarray**: hybridization-based gene expression (legacy technology)

Transcriptomics: Data & Visualization

Data Type	Visualization Task	R Tools
Differential expression	Volcano plot (log2FC vs. -log10 p-value)	EnhancedVolcano, ggplot2
Differential expression	MA plot (mean expression vs. log2FC)	DESeq2::plotMA()
Gene expression	Heatmap with hierarchical clustering	pheatmap, ComplexHeatmap
Sample quality	PCA plot	ggplot2 + prcomp()
Sample correlation	Correlation heatmap	corrplot, pheatmap
Time-series expression	Expression trend line plot	ggplot2::geom_line()
Splicing analysis	Sashimi plot (exon-junction coverage)	ggsashimi
Functional enrichment	GO / KEGG bubble plot	clusterProfiler, enrichplot
Functional enrichment	GSEA enrichment curve	enrichplot::gseaplot()
Gene set overlap	Upset plot / Venn diagram	UpSetR, ggVennDiagram
scRNA-seq	t-SNE / UMAP cell clustering	Seurat
scRNA-seq	Marker gene dot plot / violin plot	Seurat::DotPlot(), VlnPlot()
scRNA-seq	Pseudotime trajectory	monocle3
TF regulation	Gene regulatory network	igraph, ggraph

Transcriptomics: Example Figures

- **Volcano plot:** x-axis = $\log_2(\text{fold change})$, y-axis = $-\log_{10}(\text{adjusted p-value})$. Up-regulated genes on the right, down-regulated on the left. Top candidates are labeled.
- **Heatmap:** rows = genes, columns = samples, color = Z-score of expression. Row/column dendrograms show clustering. Reveals co-expressed gene modules.
- **GO/KEGG bubble plot:** x-axis = gene ratio, y-axis = pathway terms, bubble size = gene count, color = p-value. Shows which biological functions are enriched.

Section 4

Proteomics & Protein Interaction Data

What is Proteomics?

Proteomics studies the full complement of proteins — their abundance, modifications, interactions, and structures.

Typical data:

- **Mass spectrometry (MS)**: protein identification and quantification
- **Protein-protein interaction (PPI)**: physical contacts between proteins
- **Protein structure**: 3D coordinates from X-ray, cryo-EM, or prediction

Proteomics & PPI: Data & Visualization

Data Type	Visualization Task	R Tools
Protein abundance	Volcano plot (differential proteins)	ggplot2
Protein abundance	Sample correlation heatmap, PCA	pheatmap, ggplot2
PPI network	Network graph (nodes = proteins, edges = interactions)	igraph, ggraph, visNetwork
PPI network	Hub gene identification	Degree distribution plot
PPI network	Module / community detection	igraph::cluster_*
Protein domains	Domain architecture diagram	drawProteins
Protein 3D structure	Interactive 3D rendering	r3dmol, bio3d
Post-translational modifications	Modification site distribution	ggplot2::geom_segment()

Proteomics & PPI: Example Figures

- **PPI network:** a graph where highly-connected “hub” proteins often play critical roles. Communities (modules) represent protein complexes or pathways.
- **Domain architecture:** linear representation of protein domains as colored boxes along the sequence. Reveals functional modules and evolutionary conservation.
- **3D structure:** ribbon diagram showing alpha-helices and beta-sheets. Active sites and ligand binding pockets can be highlighted.

Section 5

Microbiome Data

What is Microbiomics?

Microbiomics studies communities of microorganisms (bacteria, archaea, fungi, viruses) living in a particular environment.

Typical data:

- **16S rRNA amplicon sequencing**: taxonomic profiling (genus/species level)
- **Shotgun metagenomics**: taxonomic + functional profiling
- **Metatranscriptomics**: gene expression of the community

Microbiome: Data & Visualization

Data Type	Visualization Task	R Tools
Taxonomic composition	Stacked bar plot (phylum/genus level)	ggplot2, phyloseq
Taxonomic composition	Krona / treemap (hierarchical)	ggplot2::geom_treemap()
Feature abundance	Heatmap (genus $\times$ sample)	pheatmap, geom_tile()
Alpha diversity	Box plot / violin plot by group	phyloseq, microeco
Alpha diversity	Rarefaction curve	vegan::rarecurve()
Beta diversity	PCoA / NMDS ordination	phyloseq::plot_ordination
Beta diversity	PERMANOVA / anosim results	vegan
Differential abundance	LEfSe (LDA score bar plot)	microeco
Differential abundance	Volcano plot, cladogram	ggplot2, metacoder
Phylogeny + abundance	Phylogenetic tree with abundance heatmap	ggtree + ggtreeExtra
Co-occurrence network	Microbial correlation network	igraph, SpiecEasi
Functional potential	KEGG pathway abundance heatmap	ggplot2, pheatmap
Source tracking	Sankey / alluvial diagram	ggalluvial, networkD3



Microbiome: Example Figures

- **Stacked bar plot:** each bar = one sample, colored segments = different taxa. Shows community composition at a glance. Dominant taxa and rare taxa are easily identified.
- **PCoA / NMDS:** each point = one sample, distance between points = dissimilarity in community composition. Colored by group to reveal clustering by disease status, diet, etc.
- **Phylogenetic tree + heatmap:** left panel = evolutionary tree of taxa, right panel = abundance heatmap. Reveals phylogenetic patterns in abundance.

Section 6

Epigenomics Data

What is Epigenomics?

Epigenomics studies heritable changes in gene function that do not involve changes to the DNA sequence.

Typical data:

- **ChIP-seq / CUT&RUN**: protein-DNA binding (transcription factors, histone modifications)
- **ATAC-seq**: open chromatin regions
- **Bisulfite sequencing (WGBS / RRBS)**: DNA methylation
- **Hi-C / 3C**: 3D chromatin organization

Epigenomics: Data & Visualization

Data Type	Visualization Task	R Tools
ChIP-seq / ATAC-seq	Genome browser coverage track	Gviz, ggbio
ChIP-seq / ATAC-seq	Peak annotation pie / bar chart	ChIPseeker::plotAnnoPie
ChIP-seq / ATAC-seq	TSS-centered signal heatmap	EnrichedHeatmap
ChIP-seq	Motif enrichment logo	ggseqlogo, motifStack
DNA methylation	Methylation level distribution	ggplot2::geom_violin()
DNA methylation	Differential methylation volcano	ggplot2
DNA methylation	Chromosome-level methylation profile	ggplot2
Hi-C	Contact matrix heatmap	HiCompare, pheatmap
Hi-C	TAD boundary visualization	HiCExplorer
Multi-track genome	Circos plot (integrating multiple tracks)	circlize



Epigenomics: Example Figures

- **Genome browser track:** a horizontal view of a genomic locus showing coverage peaks, annotated genes, and conservation. Tracks can be stacked for comparison across conditions.
- **TSS heatmap:** rows = genes, columns = position relative to TSS, color = ChIP signal intensity. Shows enrichment patterns at transcription start sites.
- **Contact matrix:** a symmetric heatmap where position (i, j) shows interaction frequency between genomic loci i and j . TADs appear as triangles along the diagonal.

Section 7

Clinical & Epidemiological Data

What is Clinical Bioinformatics?

Clinical bioinformatics integrates molecular data with patient clinical information to understand disease mechanisms, prognosis, and treatment response.

Typical data:

- **Survival data:** time-to-event (death, recurrence, progression)
- **Clinical covariates:** age, sex, stage, grade, treatment
- **Lab tests:** blood counts, biomarkers, imaging

Clinical Data: Data & Visualization

Data Type	Visualization Task	R Tools
Survival analysis	Kaplan-Meier survival curve	<code>survminer::ggsurvplot()</code>
Survival analysis	Cumulative hazard curve	<code>survminer</code>
Survival analysis	Cox regression forest plot (HR + CI)	<code>forestplot, ggforest</code>
Prognostic models	Nomogram (prediction tool)	<code>rms::nomogram()</code>
Prognostic models	Calibration plot	<code>rms::calibrate()</code>
Prognostic models	Time-dependent ROC curve	<code>timeROC, survivalROC</code>
Diagnostic tests	ROC curve (sensitivity vs. specificity)	<code>pROC, plotROC</code>
Diagnostic tests	Confusion matrix heatmap	<code>ggplot2::geom_tile()</code>
Risk factors	Forest plot (OR / RR with confidence intervals)	<code>forestplot</code>
Clinical features	Baseline characteristics table	<code>tableone</code>
Drug response	Dose-response curve (IC50)	<code>drc, ggplot2</code>
Multi-omics integration	Sankey / alluvial diagram	<code>ggalluvial</code>
Multi-omics integration	Cross-omics correlation heatmap	<code>corrplot,</code> <code>ComplexHeatmap</code>
Multi-omics integration	Circos / chord diagram	<code>circlize</code>



Clinical Data: Example Figures

- **Kaplan-Meier curve:** x-axis = time, y-axis = survival probability. Step-wise curves for each group. Log-rank p-value tests whether curves differ significantly.
- **Forest plot:** horizontal lines showing hazard ratio (HR) with 95% confidence interval for each variable. $HR > 1$ = increased risk, $HR < 1$ = protective effect.
- **Nomogram:** a graphical calculator where each predictor contributes points, and total points map to predicted probability. Clinicians use it for individualized risk prediction.

Section 8

The Omics Analysis Workflow

From Raw Data to Insight

Raw Data → Quality Control → Normalization → Statistical Analysis

↓

Biological Insight ← Interpretation ← Visualization

Key principle: visualization is not the final step — it is an **integral part** of every stage of analysis.

QC Visualization Examples

- **FASTQ quality scores:** per-base quality box plots
- **Alignment statistics:** mapping rate, duplication rate bar charts
- **Sample correlation:** PCA or correlation heatmap to detect outliers
- **Library complexity:** rarefaction curves for microbiome

Analysis Visualization Examples

- **Differential expression:** volcano plot, MA plot
- **Clustering:** heatmap, PCA, t-SNE/UMAP
- **Enrichment:** bubble plot, enrichment map, cnetplot
- **Classification:** ROC curve, confusion matrix

Interpretation

Good figures tell a story:

- What is the **biological question**?
- What does the figure **show**?
- What is the **take-home message**?

Section 9

Summary

What We Covered Today

Omics Type	Key Data	Typical Visualizations
Genomics	Mutations, CNV, GWAS	Oncoplot, Manhattan, Circos
Transcriptomics	RNA-seq, scRNA-seq	Volcano, Heatmap, t-SNE/UMAP
Proteomics & PPI	Protein abundance, interactions	Network graph, 3D structure
Microbiome	16S rRNA, metagenomics	Stacked bar, PCoA, Heatmap
Epigenomics	ChIP-seq, methylation, Hi-C	Genome browser, Contact matrix
Clinical	Survival, risk factors	KM curve, Forest plot, Nomogram

Key Takeaway

Every omics data type has its own characteristic data structures and visualization needs. Understanding what visualizations are appropriate for each data type is essential for bioinformatics research.

Looking Ahead

In future courses, we will work through each omics data type with **real-world datasets**, covering:

- ① Data loading and preprocessing
- ② Statistical analysis
- ③ Visualization with R
- ④ Results interpretation

Next Talk

Talk 08: Strings, Iteration & Functional Programming (`stringr` + `purrr` + `apply` family)

No Homework

This talk is an overview — no homework assignment. See you in Talk 08!